A Sample Article Using IEEEtran.cls for IEEE Journals and Transactions

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***Abstract*—This document describes the most common article elements and how to use the IEEEtran class with LATEX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.**

***Index Terms*—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.**

I. Introduction

THIS file is intended to serve as a "sample article file"

II. Related work

A. Hyperbola recognition task

In GPR B-scan images, buried pipes, cables, and roots typically appear as hyperbolic reflections, making hyperbola detection a prerequisite for target localization, depth estimation, and parameter reconstruction [1]–[3]. Early methods relied on hand-crafted features and analytical fitting, screening candidates with neural preprocessing, Viola–Jones, or HOG+SVM and then localizing them via the Hough transform or template/least-squares fitting [3]–[7]. Such pipelines work in constrained settings but remain computationally expensive, sensitive to parameter discretization, and dependent on prior template design or user intervention [2], [6], [8], [9]; later refinements combined edge detection and C3 clustering with Hough fitting to improve robustness [9], [10].

Deep learning shifted the field from sliding-window recognition toward end-to-end detection on full B-scans. Two-stage detectors led this transition: Faster R-CNN, trained with simulated radargrams to offset scarce field labels [8], was later coupled with explicit hyperbola fitting for automatic localization [11], while a CNN-LSTM reformulated diameter identification as region classification and reached 92.5% accuracy on field data [2]. One-stage detectors emphasized real-time deployment, with YOLOv3 running at about 12 fps [12], combined with migration-based focusing and apex extraction to estimate pipeline depth within 0.04 m [13], and a YOLOv4 variant tailored to root recognition and localization [14].

More recent work argues that a generic axis-aligned box is suboptimal for a structured geometric echo, a limitation inherent to mainstream detectors that regress box bounds or centers [15], [16]. GPR-specific methods therefore inject geometric priors: anchor-optimized Mask Scoring R-CNN [17], GPR-Former, which directly regresses hyperbola parameters and refines them with a symmetry constraint [1], and shape-aware topological representations fused with YOLOv5 [18]. Overall, the field has progressed from hand-crafted curve search to geometry-aware detection, yet most detectors still localize hyperbolas through rectangular proxies rather than modeling their strip-like, symmetric, parametric shape explicitly.

B. Attention Mechanisms in GPR Hyperbola Interpretation

Attention mechanisms are increasingly embedded into task-specific GPR pipelines so that models respond to hyperbola-relevant structures rather than processing the whole radargram uniformly. Localized hyperbolic attention enhances hyperbolic wave features and suppresses clutter to improve downstream imaging of target pipelines [19], while a dual-cascade network first removes direct-wave interference and then applies attention with feature fusion to raise hyperbola detection accuracy before geometric localization [20]. Because hyperbolic reflections extend across neighbouring traces and the full time window, transformer-style self-attention is also used to model the long-range dependencies that convolution captures less efficiently: GPR-Former regresses hyperbola parameters directly from B-scans and refines them with a symmetry constraint [1], and GPR-TransUNet introduces self-attention into a TransUNet-style regression framework for the same reason [21].

More recent designs tailor attention specifically to hyperbolic geometry and apex-localization objectives that are closely aligned with our study. A rebar-recognition model emphasizes hyperbolic features with deformable attention and a hyperbolic attention transformer before proposal generation [22], a lightweight dual-task keypoint detector couples attention-enhanced features with keypoint regression for precise vertex localization [23],

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and a state-space detector for arbitrarily long B-scans targets hyperbola vertex estimation directly [24]. The clear trend is a shift from generic channel or spatial reweighting toward physically informed designs that encode hyperbolic structure and apex localization, while traditional fitting-based pipelines remain relevant for their geometric interpretability [11]. This motivates our approach: instead of localizing the echo through rectangular proxies or post-hoc curve fitting, we represent the hyperbola by its strip-like, symmetric, parametric shape, so that both annotation and learning are aligned with the geometry of the target.

C. Labelling tools

Supervised detectors depend on large sets of manually annotated images, and a range of general-purpose annotation tools has been developed to meet this need. Early web-based systems such as LabelMe established collaborative image labelling at scale [25], and the annotation pipeline behind large benchmarks like MS-COCO further standardized bounding-box, polygon, and segmentation-mask labelling [26]. Building on these efforts, widely used open-source tools including LabelImg [27], Labelme [28], the VGG Image Annotator [29], and CVAT [30] provide interactive interfaces for the common geometric primitives—axis-aligned rectangles, polygons, key points, and pixel-wise masks. These primitives are generic and domain-agnostic, which is exactly what makes the tools broadly applicable across natural-image tasks.

In GPR interpretation, hyperbolic reflections are usually annotated with these same general-purpose tools; bounding boxes drawn in software such as LabelImg are the de facto standard for labelling subsurface targets in B-scans [31]. A rectangle, however, records only the extent of the echo rather than its strip-like, symmetric, and parametric geometry, so the resulting labels are misaligned with the physical shape of the hyperbola and cannot directly provide geometric supervision such as vertex position, curvature, or band thickness. Polygon and mask tools can trace the curve more tightly, but they require dense per-instance clicking and still do not expose the underlying hyperbola parameters. This gap between the geometry-agnostic primitives offered by existing tools and the parametric, band-shaped nature of hyperbolic targets motivates the dedicated annotation tool proposed in this work, which lets annotators place and adjust a parametric hyperbola band through a few interpretable controls and export both the shape parameters and the derived bounding box.

III. Proposed method

A. Hyperbola annotation

We designed a visual annotation tool for hyperbolic ribbon-shaped targets, whose user interface is illustrated in the Fig. 1. First, the user selects an image folder, and the tool automatically loads all images within it. The original image is displayed on the lower-left panel, while the annotated image is shown on the right. A mouse click on the original image marks the vertex position of the hyperbola, and five adjustable parameters are provided on the right panel.

The five parameters are defined as follows: the horizontal and vertical coordinates of the hyperbola vertex, hyperbola width, height, and ribbon thickness. By clicking the *Add as New Annotation* button, the current parameter configuration is saved as a new annotation for the active image, and multiple annotations can be added to a single image.

All annotations are stored in JSON format with the following structure:

```
{
  "024.jpg": [
    {
      "label": "hyperbola",
      "x_vertex": 127.04,
      "y_vertex": 163.84,
      "width": 87.65,
      "height": 59.76,
      "thickness": 33.03
    },
    {
      "label": "hyperbola",
      "x_vertex": 186.56,
      "y_vertex": 109.76,
      "width": 97.35,
      "height": 79.35,
      "thickness": 33.03
    }
  ]
}
```

Here, *024.jpg* denotes the image filename, and the remaining entries represent the parameters and their corresponding values.

After the hyperbola band is annotated, the conventional bounding-box annotation can be directly derived from the boundary of the band. Fig. 2 shows an example, presenting the raw image, the annotated hyperbola band, and the corresponding bounding-box form.

B. Hyperbola-attention nerual network

The network takes a single-channel grayscale B-scan image as input and outputs a rectangular bounding

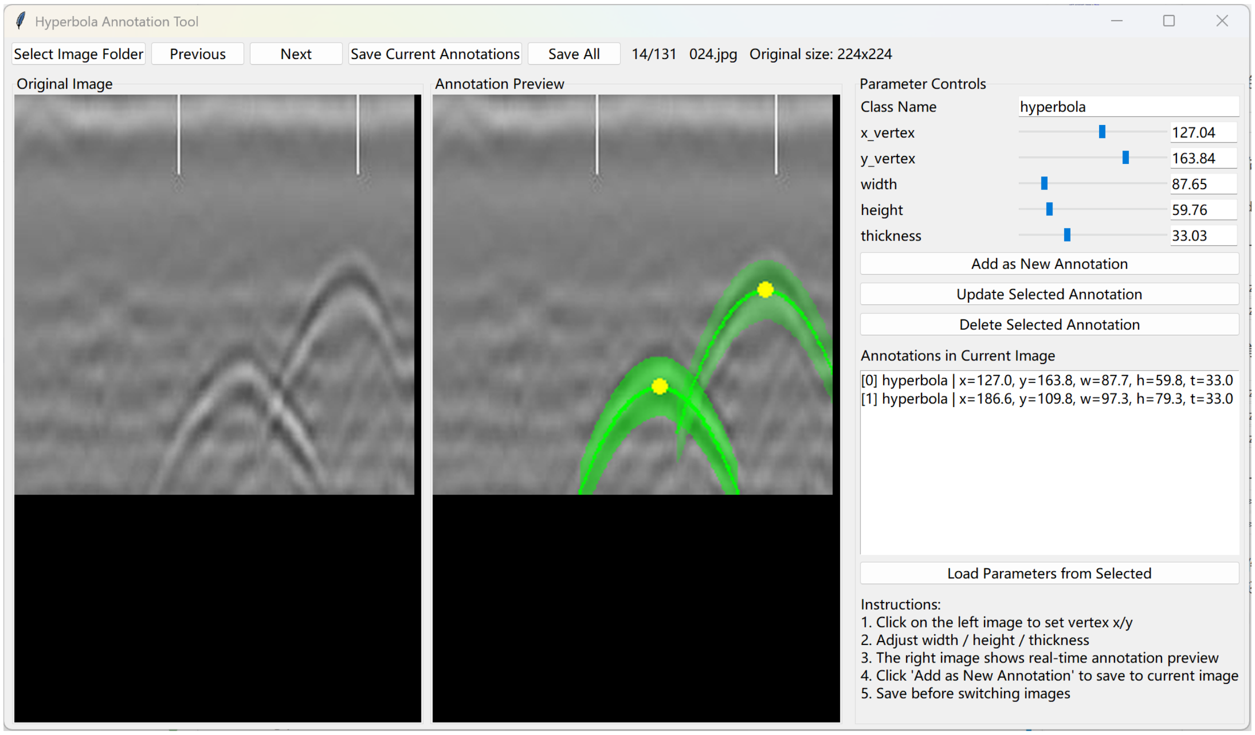
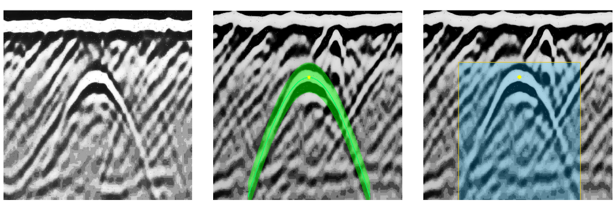


Fig. 1. tool



(a) Raw image. (b) Annotated hyperbola band. (c) Derived bounding box.

Fig. 2. Example of a hyperbola sample: the raw image, the annotated hyperbola band, and the bounding box derived from the band boundary.

box for each hyperbolic signature in the image. Unlike existing methods, we introduce within the backbone an attention branch that is explicitly supervised by a hyperbola-band mask, and inject it into the bottleneck features via channel-wise concatenation, thereby guiding the network to focus on the hyperbolas and improving detection performance. A schematic of the network is shown in Fig. 3.

The backbone consists of three down-sampling stages followed by a bottleneck layer; each stage comprises two 3×3 convolutions, each followed by batch normalization and ReLU, together with a subsequent 2×2 max-pooling operation that doubles the channel depth while halving the spatial resolution. Across the three stages the resulting feature maps are 1×640^2, 32×320^2, 64×160^2, and 128×160^2, further reduced to 256×80^2 at the bottleneck. The attention branch takes f_3 as input rather than the bottleneck feature F_b, as f_3 retains a higher spatial resolution that is better suited to precise localization.

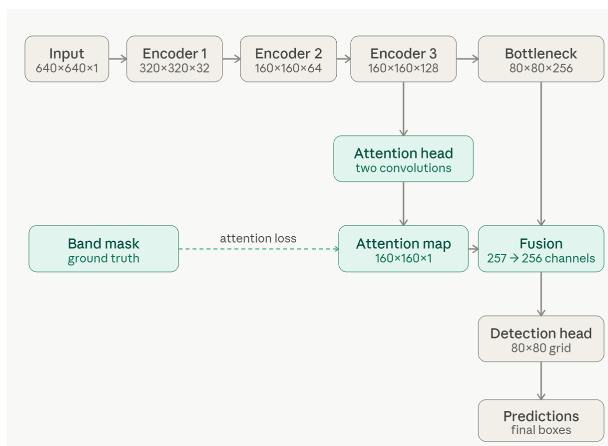


Fig. 3. 11

The supervised attention branch is the central component of the proposed method. A lightweight head $g_a(\cdot)$, consisting of a 3×3 convolution with batch normalization and ReLU followed by a 1×1 convolution, maps f_3 to a single-channel logit that is passed through a sigmoid to yield the attention map A. The supervisory target B is a band mask rasterized from the annotated hyperbola geometry. The attention map is then supervised pixel-wise to approximate B using a combined binary cross-entropy and Dice loss, $\mathcal{L}_{\mathrm{att}} = \mathcal{L}_{\mathrm{bce}} + \mathcal{L}_{\mathrm{dice}}$. Binary cross-entropy can be understood as an independent yes/no classification at every pixel, penalizing each pixel that is misclassified as belonging to, or not belonging to, the hyperbola band. The Dice term instead measures the overall overlap between the predicted and ground-truth regions, with the loss decreasing as the overlap increases. The two terms are complementary: the former enforces pixel-level accuracy, while the latter enforces region-level shape consistency and helps mitigate the class imbalance caused by the band occupying only a small fraction of the image.

a) Attention-guided feature fusion.: The attention map A is passed through a sigmoid and downsampled by a factor of two via average pooling to align with the bottleneck resolution, yielding $\tilde{A}$. It is then concatenated with F_b along the channel dimension and compressed back to 256 channels by a 1×1 convolution with batch normalization and ReLU, producing the fused feature $\hat{F}$. This operation injects the spatial prior of the hyperbola band into the feature representation passed to the detection head.

b) Detection head and decoding.: The detection head follows a YOLO-style dense prediction paradigm, producing three parallel outputs over an 80×80 grid at stride 8: objectness (1 channel), box size (2 channels), and sub-pixel center offset (2 channels). During inference, grid cells whose sigmoid-activated objectness exceeds a threshold τ are decoded into bounding boxes, and redundant detections are removed by box-level IoU-based non-maximum suppression. The head is trained with a detection loss $\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{obj}} + \mathcal{L}_{\mathrm{size}} + \mathcal{L}_{\mathrm{off}}$, aggregating the objectness, box-size, and center-offset terms, and the overall training objective combines this with the attention supervision loss introduced above,

$$\mathcal{L} = \mathcal{L}_{\mathrm{det}} + \lambda_{\mathrm{att}}\,\mathcal{L}_{\mathrm{att}}, \tag{1}$$

where λ_{att} controls the relative weight of the attention supervision. All terms are optimized jointly in a single end-to-end training pass.

IV. Experiment

V. Result and evaluation

A. Dataset

The experiments are conducted on a publicly available GPR dataset for intelligent recognition of subsurface utilities [32]. The dataset consists of labeled GPR B-scan images collected from real urban and industrial environments using ground-coupled antennas operating at 200 MHz and 400 MHz.

B. Detection performance

Table I reports the detection performance under different training-data ratios. As can be observed, the proposed method consistently outperforms the baseline across all training-data ratios. The improvement in F1-score is most pronounced at a training-data ratio of 0.7, where our method achieves 0.84 compared with 0.68 for the baseline. The training stability is also notably enhanced, with the standard deviation decreasing from 0.21 to 0.03. This gain is primarily attributed to the improvement in recall, as the attention constraint guides the network to focus on the hyperbola-band regions and thereby reduces missed detections, which demonstrates the effectiveness of the proposed method.

Nevertheless, at a training-data ratio of 0.9, no significant improvement is observed and the stability is relatively poor. This is presumably due to the limited number of samples in the test set, which leads to unstable evaluation results.

It should be noted that the above results are obtained with $\lambda_{\mathrm{att}} = 0.7$; a sensitivity analysis of the parameter λ is further presented in the following section.

C. Comparison of Detection Performance with Different Attention Modules

As shown in Table II, we compare the proposed method with several representative attention modules that have been applied to ground-penetrating radar, SAR, and remote-sensing recognition: Pan et al. adopt the channel attention SE, N. Wang et al. adopt the channel–spatial attention CBAM, Z. Wang et al. adopt the non-local attention Non-local, and Shen et al. adopt the coordinate attention CA. For a fair comparison, we re-implement these attention modules at the same position within an identical backbone and evaluate them on our hyperbola dataset under exactly the same training and evaluation settings, so that all results reported in the table are obtained by our own re-implementation. The proposed method achieves the best precision, recall, and F1, reaching an F1 of 0.89 with an improvement of about 9% over the second-best method. Whereas the existing modules generally struggle to balance precision and recall, our method attains the highest precision and recall simultaneously, together with the lowest F1 standard deviation, thus offering both higher detection accuracy and greater stability. This can be attributed to the fact that SE, CBAM, Non-local, and CA are all self-learned attention mechanisms that merely re-weight channel or spatial features according to the features themselves, without explicit guidance on where to attend; in contrast, our method imposes explicit supervision on the attention branch through the hyperbola-band mask, directly guiding the network to focus on the hyperbolic

TABLE I
DETECTION PERFORMANCE (MEAN ± STD OVER 10 SEEDS) AT DIFFERENT TRAINING-DATA RATIOS. "OURS" USES THE ATTENTION CONSTRAINT WITH $\lambda_{\mathrm{att}} = 0.7$. THE BEST F1 PER ROW IS SHOWN IN **BOLD**.

Training ratio	Baseline			Ours		
	Precision	Recall	F1-score	Precision	Recall	F1-score
0.5	0.81 ± 0.20	0.43 ± 0.09	0.53 ± 0.08	0.86 ± 0.08	0.49 ± 0.07	$\mathbf{0.62 \pm 0.06}$
0.6	0.71 ± 0.21	0.56 ± 0.08	0.60 ± 0.10	0.75 ± 0.25	0.65 ± 0.06	$\mathbf{0.66 \pm 0.13}$
0.7	0.75 ± 0.30	0.68 ± 0.10	0.68 ± 0.21	0.91 ± 0.08	0.78 ± 0.05	$\mathbf{0.84 \pm 0.03}$
0.8	0.84 ± 0.22	0.82 ± 0.07	0.81 ± 0.16	0.91 ± 0.12	0.88 ± 0.04	$\mathbf{0.89 \pm 0.07}$
0.9	0.88 ± 0.24	0.90 ± 0.07	0.87 ± 0.17	0.85 ± 0.22	0.93 ± 0.04	$\mathbf{0.87 \pm 0.17}$

TABLE II
COMPARISON OF DETECTION PERFORMANCE WITH DIFFERENT ATTENTION MODULES

Method	Precision	Recall	F1-score
Pan et al. [33]	0.86 ± 0.20	0.82 ± 0.04	0.82 ± 0.11
N. Wang et al. [31]	0.75 ± 0.17	0.82 ± 0.03	0.77 ± 0.10
Z. Wang et al. [34]	0.84 ± 0.28	0.74 ± 0.23	0.78 ± 0.25
Shen et al. [35]	0.80 ± 0.23	0.84 ± 0.09	0.80 ± 0.17
Ours	$\mathbf{0.91 \pm 0.12}$	$\mathbf{0.88 \pm 0.04}$	$\mathbf{0.89 \pm 0.07}$

target regions and thereby outperforming these generic attention modules.

D. Result3

E. Result4

F. Result3

G. Sensitivity analysis of the parameter λ

Fig. 4 illustrates the distribution of the detection F1 as the attention-constraint weight λ varies from 0.1 to 10, where each value is evaluated over 10 random seeds and $\lambda = 0$ corresponds to the baseline. Across the entire range of λ, the median F1 obtained with the attention constraint consistently remains between 0.78 and 0.84, substantially higher than the baseline value of 0.68. This indicates that the proposed method is insensitive to the choice of λ and yields stable performance gains over a wide range of this parameter. The best result is achieved at $\lambda = 0.7$, which attains the highest median F1 of 0.84 together with the most concentrated distribution, with a standard deviation of only 0.03. When λ is too small, the constraint becomes too weak to provide sufficient guidance, whereas an excessively large λ slightly suppresses the primary detection task and lowers the F1 to around 0.79. These results suggest the existence of an intermediate optimal range, and $\lambda = 0.7$ is therefore adopted as the default setting throughout our experiments in consideration of both accuracy and stability.

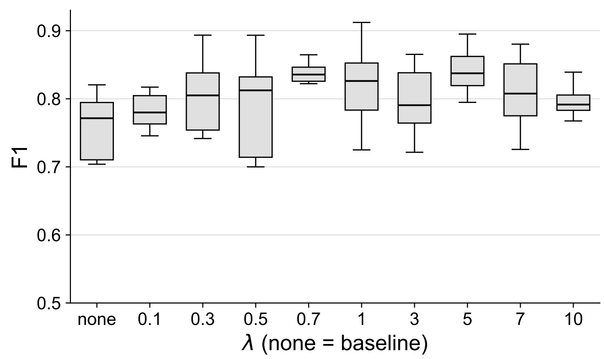


Fig. 4. lambda

H. Result3

VI. DISCUSSION

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This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

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